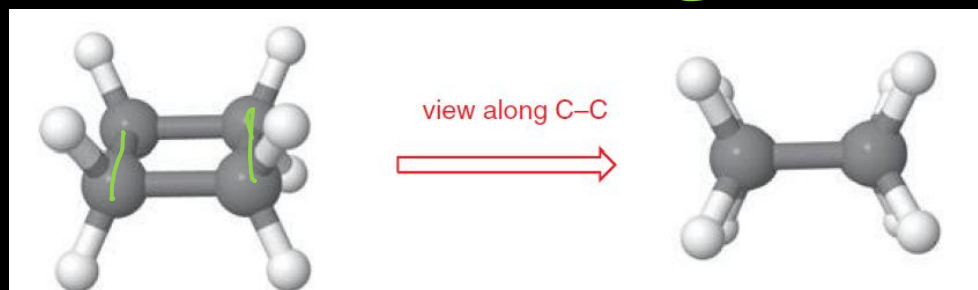
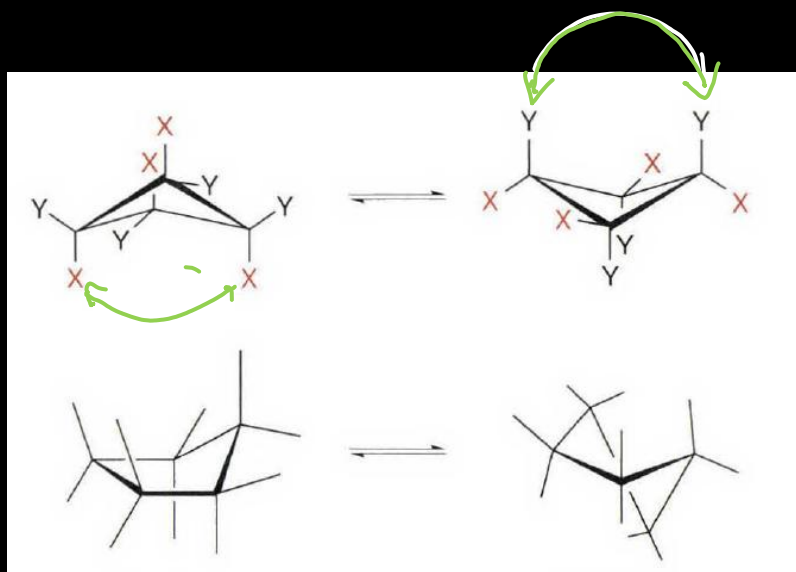
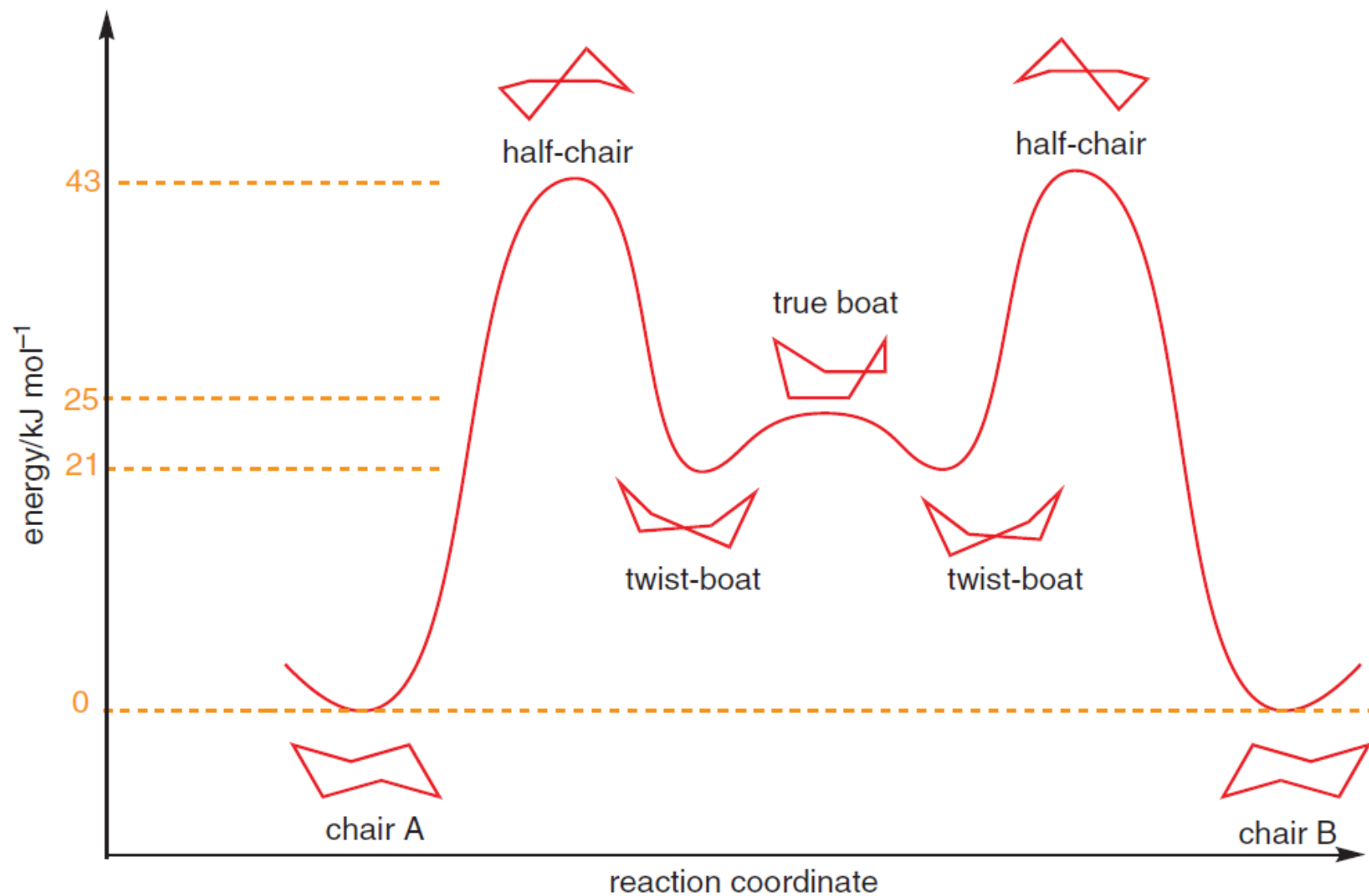
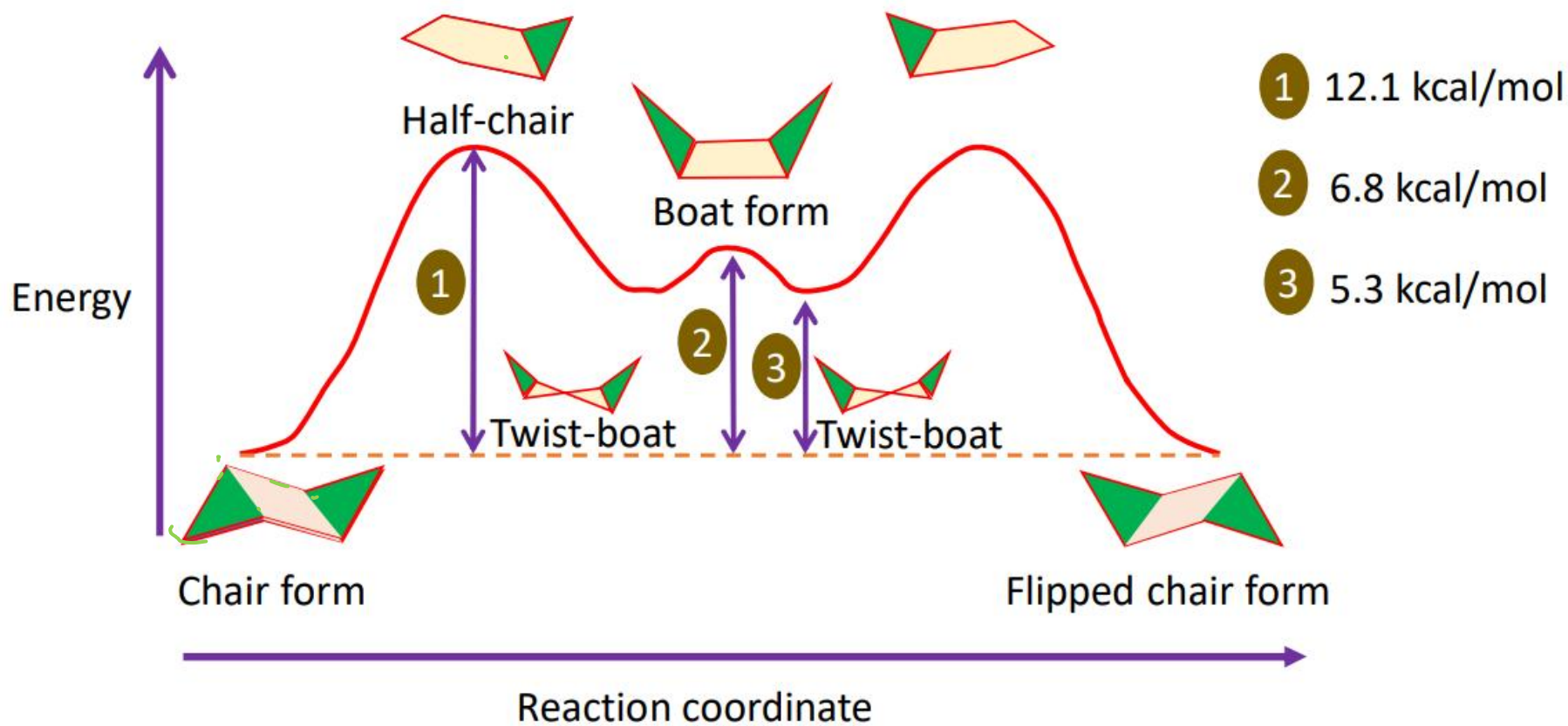
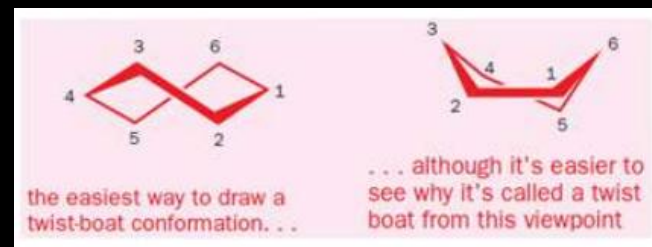
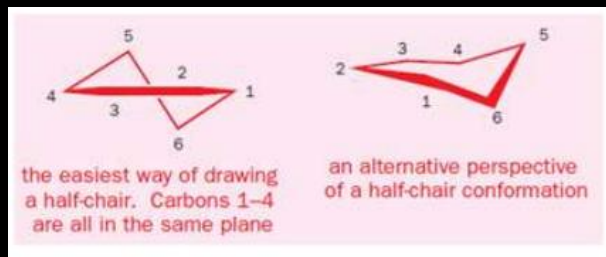


Cyclopropane



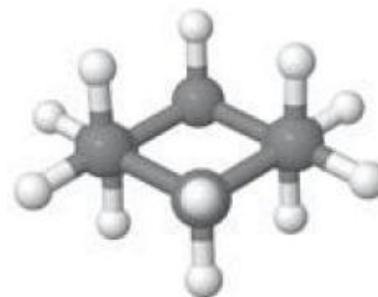
conformational changes during the inversion of cyclohexane



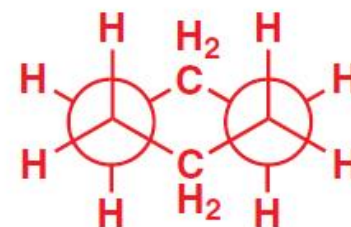




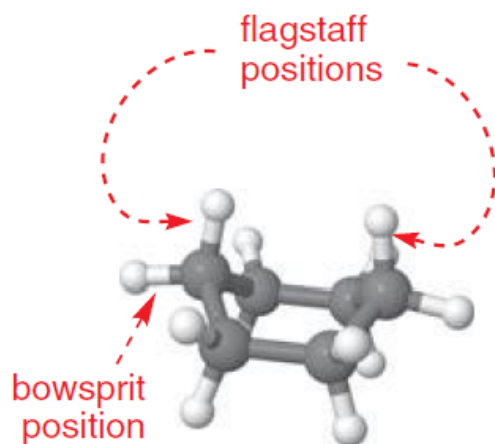
a side-on view of the chair conformation of cyclohexane



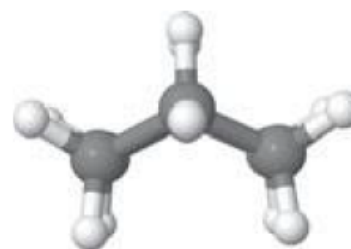
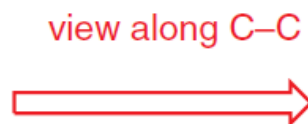
a view of cyclohexane looking along two of the C—C bonds



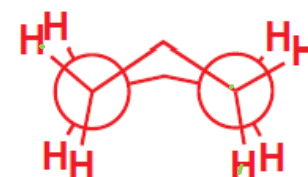
a Newman projection of the same view



a side-on view of the boat conformation of cyclohexane

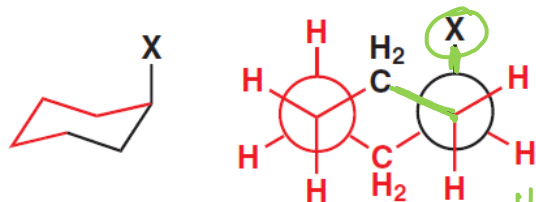


a view of the boat conformation looking along two of the C—C bonds



Newman projection of the same view

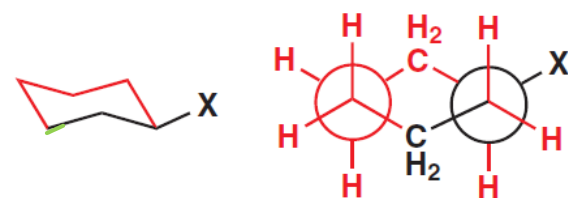
axially substituted cyclohexane:



the black bonds are synclinal (gauche)
(only one pair shown for clarity)

Axial
1,3-Diaxial
interaction

equatorially substituted cyclohexane:

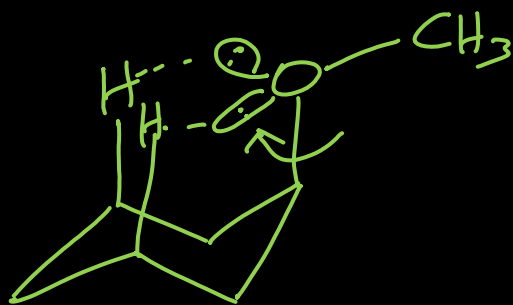


the black bonds are anti-periplanar
(only one pair shown for clarity)



$$K = \frac{\text{concentration of equatorial conformer}}{\text{concentration of axial conformer}}$$

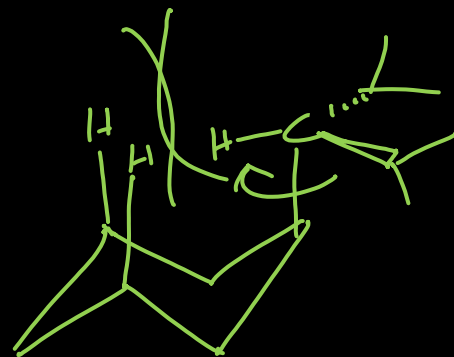
X	Equilibrium constant, K	Energy difference between axial and equatorial conformers, kJ mol^{-1}	Percentage with substituent equatorial
H	1	0	50
OMe	2.7	2.5	73
Me	19	7.3	95
Et	20	7.5	95
<i>i</i> -Pr	42	9.3	98
<i>t</i> -Bu	>3000	>20	>99.9
Ph	110	11.7	99



27%



5%

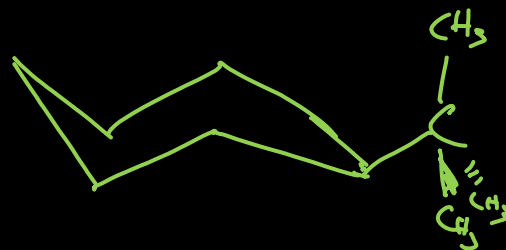


2%



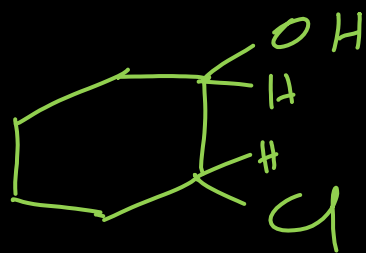
Strong 1,3-diaxial interaction

~0.01%



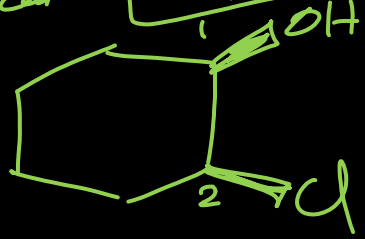
99.9%

(*t*-Bu $\Rightarrow$ locks the eq. orientation)
Bulky groups $\Rightarrow$ Shifts the equilibrium towards equatorial orientation



Geometrical isomers
 $\Rightarrow$

1,2 $\rightarrow$ cis
 eq & ax

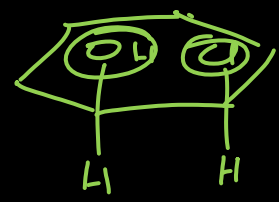


cis

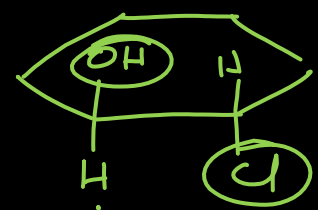
1,2-trans
 ax-ax / eq-eq



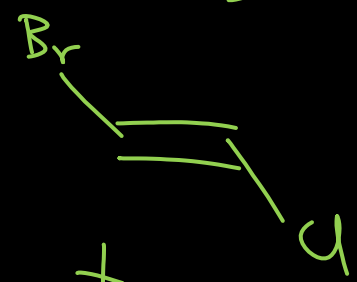
trans



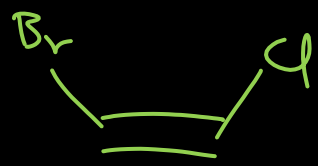
cis



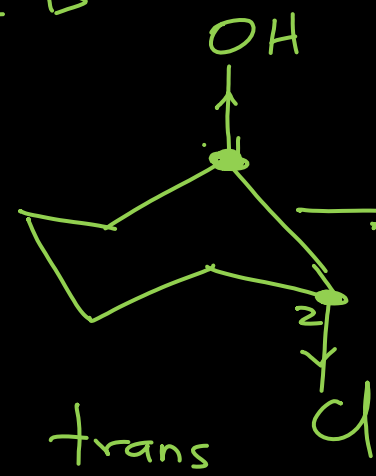
trans



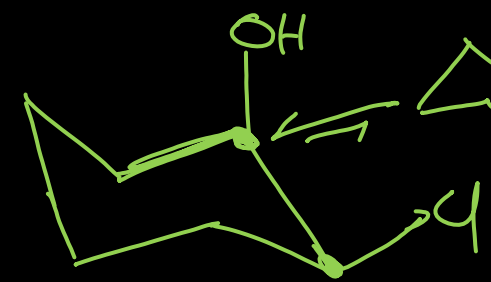
trans



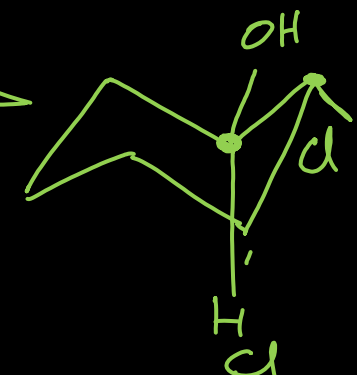
cis

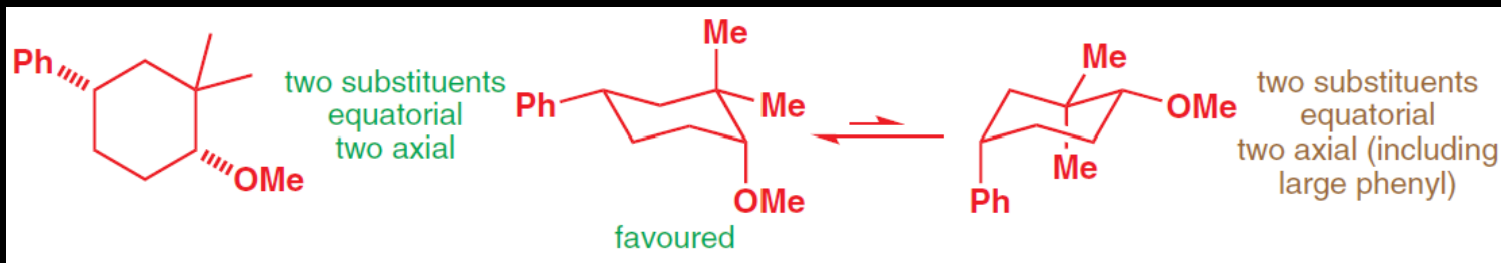
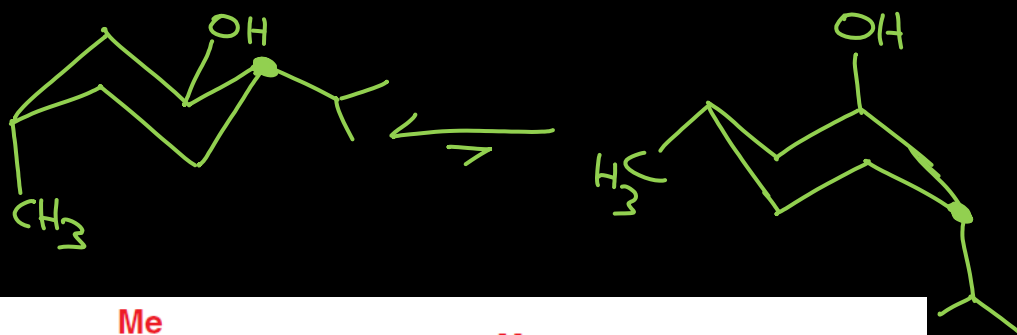
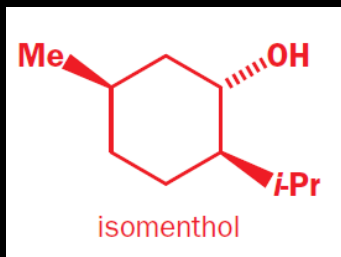
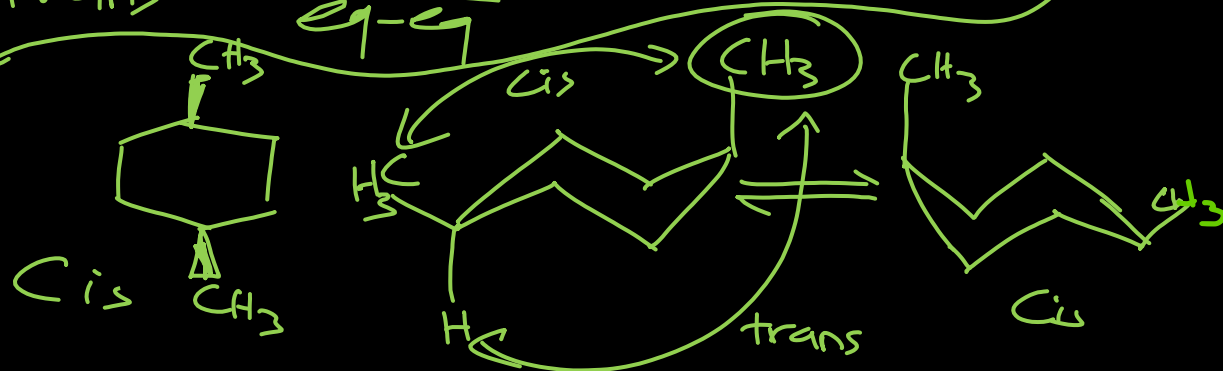
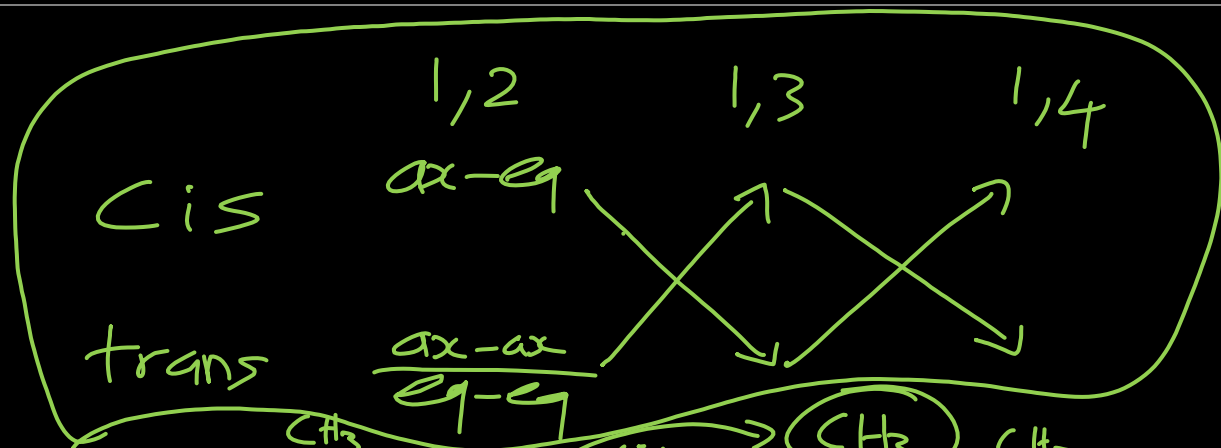
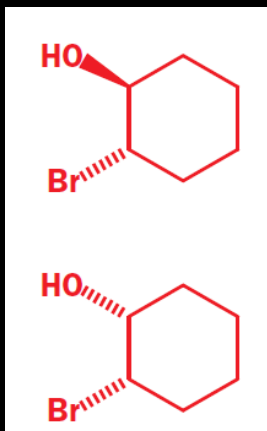


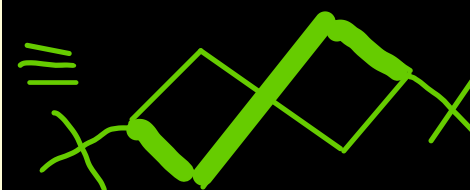
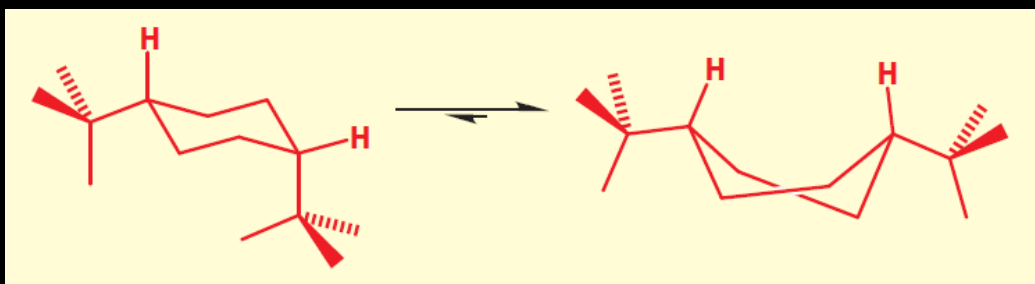
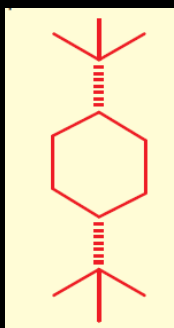
trans



cis







Favours
Twist boat conformer

